Sun Belt Football Spending & Performance Analysis

Python Companion Analysis Summary

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Project objective

This Python companion analysis reproduces and extends an Excel-based project comparing 2023 Sun Belt football performance with EADA Survey Year 2024 reported football expenses across 14 NCAA Division I programs.

Python workflow

Step	Description
1. Data import	Loaded the cleaned Excel dataset used in the dashboard.
2. Data preparation	Standardized columns and selected the variables required for analysis.
3. KPI calculation	Recalculated win percentages, expenses per participant, wins per $1M, and conference wins per $1M.
4. Validation	Compared Python-calculated metrics with the Excel workbook outputs.
5. Analysis	Ranked programs by spending efficiency and analyzed correlations between spending and performance.
6. Visualization	Generated scatter plots and spending-efficiency bar charts using matplotlib.

Key Python outputs

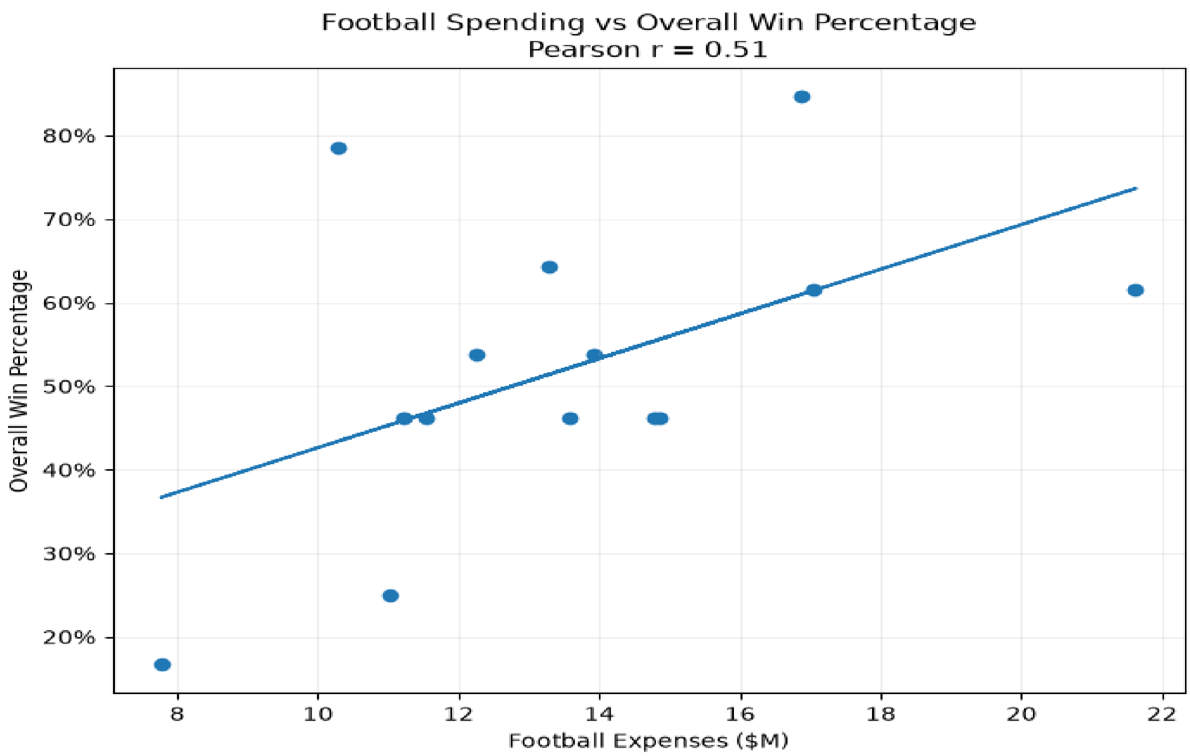
Metric	Result
Correlation: expenses vs overall win percentage	Pearson r = 0.51
Correlation: expenses vs conference win percentage	Pearson r = 0.41
Explained linear variation for conference win percentage	R-squared approximately 0.16
Top spending-efficiency performer by wins per $1M	Troy University
Strong performance at high spending level	James Madison University

Interpretation

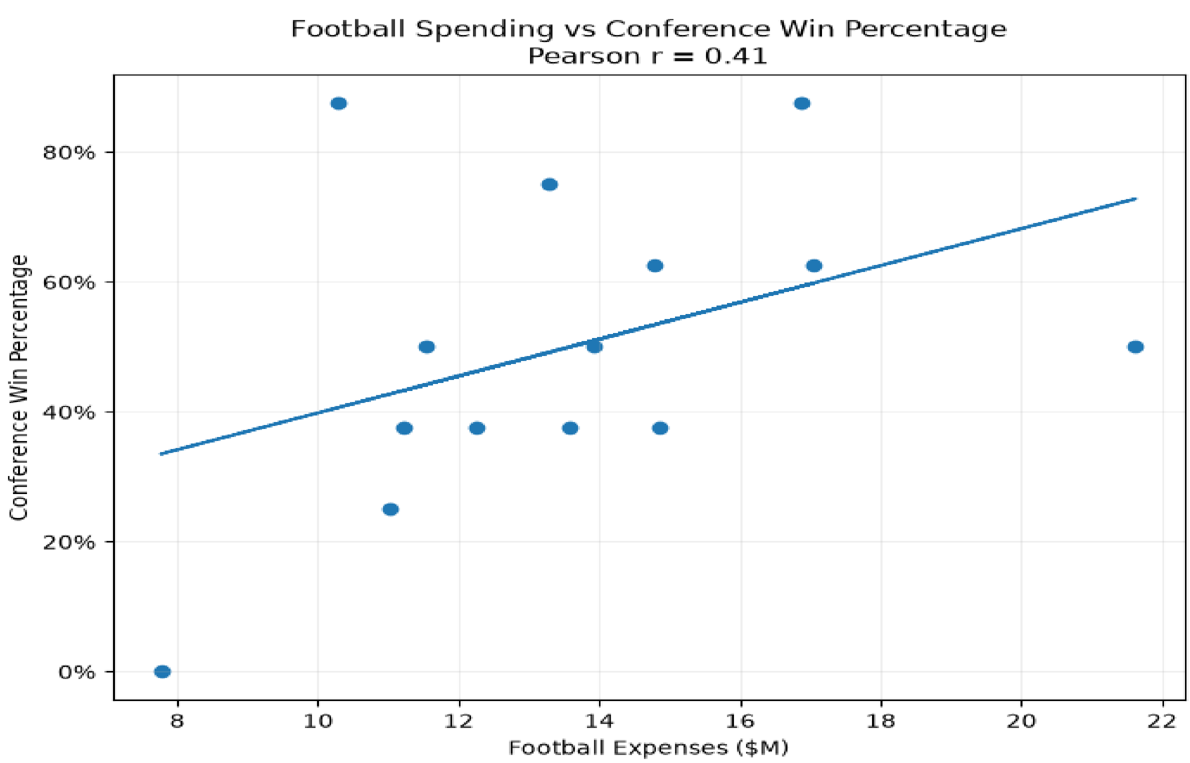
The results suggest a moderate positive relationship between reported football expenses and performance, but the relationship is limited. Spending alone does not explain most of the observed variation in conference win percentage, and the analysis is descriptive rather than causal.

Python-generated visualizations

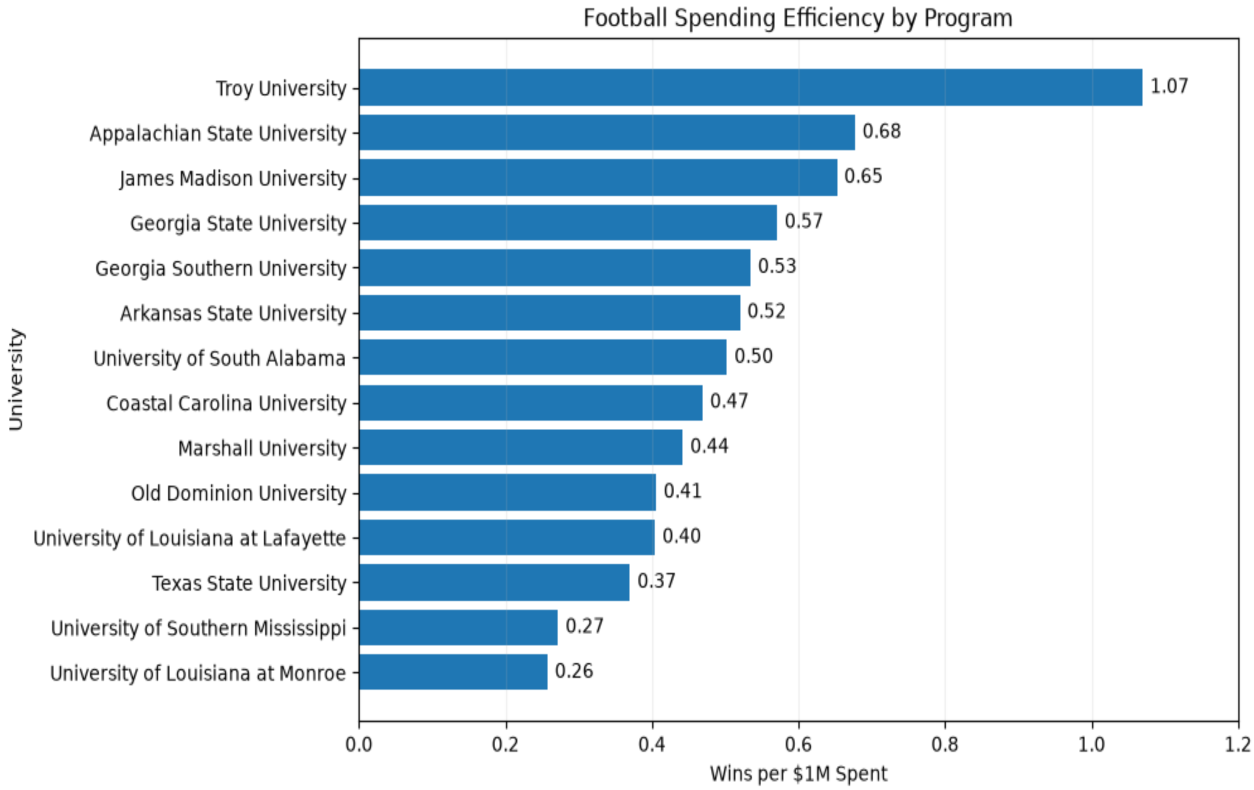
Football spending vs overall win percentage



Football spending vs conference win percentage



Football spending efficiency by program



Limitations

- The analysis covers one conference and one season only.

- Correlation does not establish causation.

- Strength of schedule, injuries, coaching changes, roster quality, and recruiting strength are not controlled.

- EADA expenses exclude capital expenditures, debt service, and certain indirect facility costs.

- EADA reported expenses are annual actual expenditures, not a preseason budget.

- Overall-season metrics are less directly comparable because teams played different numbers of games due to postseason participation.

Files

The full project includes the Excel workbook, raw EADA CSV export, Python/Jupyter notebook, project PDF dashboard, and this Python companion summary.

Repository-ready structure

data/ for source files, notebooks/ for the Python notebook, outputs/ for the dashboard PDF and rendered notebook outputs, README.md for project documentation.